Name _____ **Date** _____

Worksheet 3.1

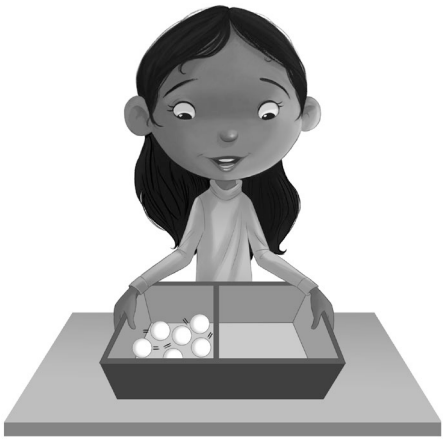
Make a particle model

You will need: a rectangular box such as a shoe box, a piece of cardboard that fits the width of the box to divide it into sections, six marbles.

Follow these steps:

Step 1

- Put the divider about halfway along the bottom of the box to decrease the space for the marbles. Put the marbles in one section only. Shake the box in all directions, but without lifting the box from the table.
- Observe the movement of the marbles.
- Observe the spaces between the marbles.



Step 2

- Put the divider close to one end of the box to decrease the space for the marbles even more. This time the marbles must be as closely packed as possible. Shake the box in all directions, but without lifting the box from the table.

- Observe the movement of the marbles.
  How is the movement different to the movement of the marbles in steps 1 and 2?

  _____

- Observe the spaces between the marbles.
  How are the spaces different to the spaces between the marbles in steps 1 and 2?

  _____

Questions

1 You have made two arrangements with the box and marbles.

a Which one do you think is a model for the particles in a solid?

_____

b Which do you think is a model for the particles in a liquid?

_____

c Say why you chose these arrangements for particles in a solid and a liquid.

_____

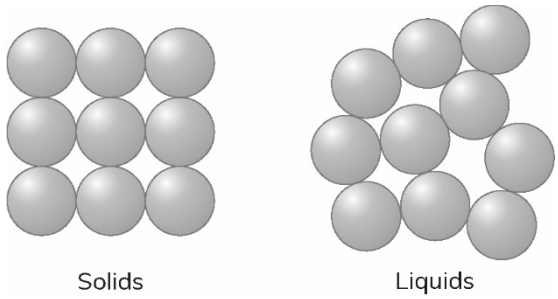
_____

_____

_____

Help sheet

The drawings show the arrangement of particles in solids and liquids.

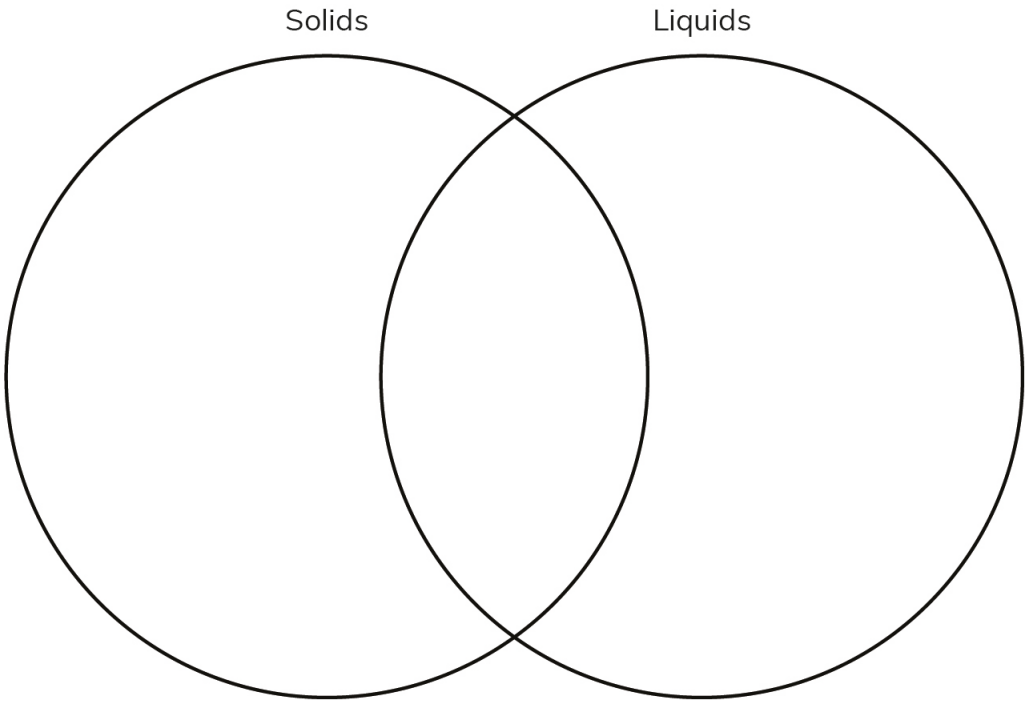


Stretch sheet

2 What else could you have used to represent the particles of matter?

_____

3 Fill in the Venn diagram to compare the arrangement and movement of particles in solids and liquids.



Name _____ **Date** _____

Worksheet 3.2

Make a sand clock

A sand timer is used to measure time. Sand flows from one glass bulb through a narrow neck into another glass bulb below.

Different sand timers are made to measure different lengths of time. For example, in sand timers that we use to time how long eggs must boil, it takes about four minutes for all the sand to flow from the top bulb into the bottom bulb. Different factors affect how fast the sand flows.

You are going to make your own sand timer.

You will need: two jars or bottles of the same type, such as soft drink bottles or jam jars, sand, aluminum foil, electrical tape or duct tape, a timer, a sharp pencil.

- Decide in your group how you will use the materials to make your sand timer.
- Make the sand timer and time how long it takes for all of the sand to flow out of the top bottle.

Questions

1 How long did it take for all of the sand to flow out of the top bottle?

_____

2 Name **two** factors that can affect how fast the sand flows.

_____

_____

3 a Suggest how you could increase the amount of time it takes for the sand to flow from one bottle in to the other bottle.

_____

_____

b How could you shorten the time?

_____

Help sheet

- Half-fill one bottle with sand. Cover the top of the bottle with the small piece of aluminium foil, shape it to fit the bottle and tape it in place.
- Make a hole through the middle of the piece of aluminium foil with a sharp pencil. Check that it is large enough for sand to flow freely.
- Place the second and empty bottle on top of the first, neck-to-neck. Put tape around the necks of the two bottles to hold them together.

Stretch sheet

4 How can you make the timer measure a certain amount of time, such as the time for hardboiling an egg? Think of a way and try it.

_____

_____

5 Would this be a good clock to try to tell the time with? Say why or why not.

_____

_____

Name _____ **Date** _____

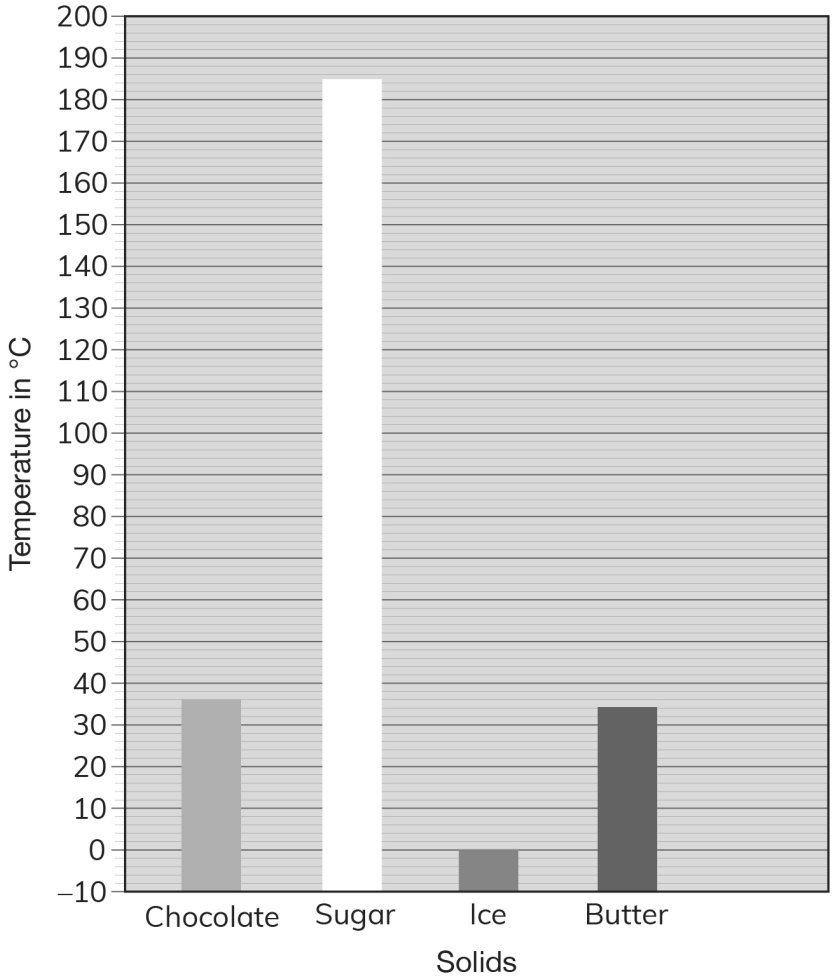
Worksheet 3.3A

Compare melting in different solids

'Melting point' is the temperature at which a solid melts.

Different solids have different melting points.

Look at the bar chart, which shows the melting points of four solids.



1 What unit is used to measure melting point on this graph?

_____

2 a Which solid has the highest melting point?

_____

b What is the melting point of this solid?

_____

3 a Which solid has the lowest melting point?

_____

b What is the melting point of this solid?

_____

4 Beeswax melts at 62 °C. Draw a bar on the graph to show this.

Name _____ **Date** _____

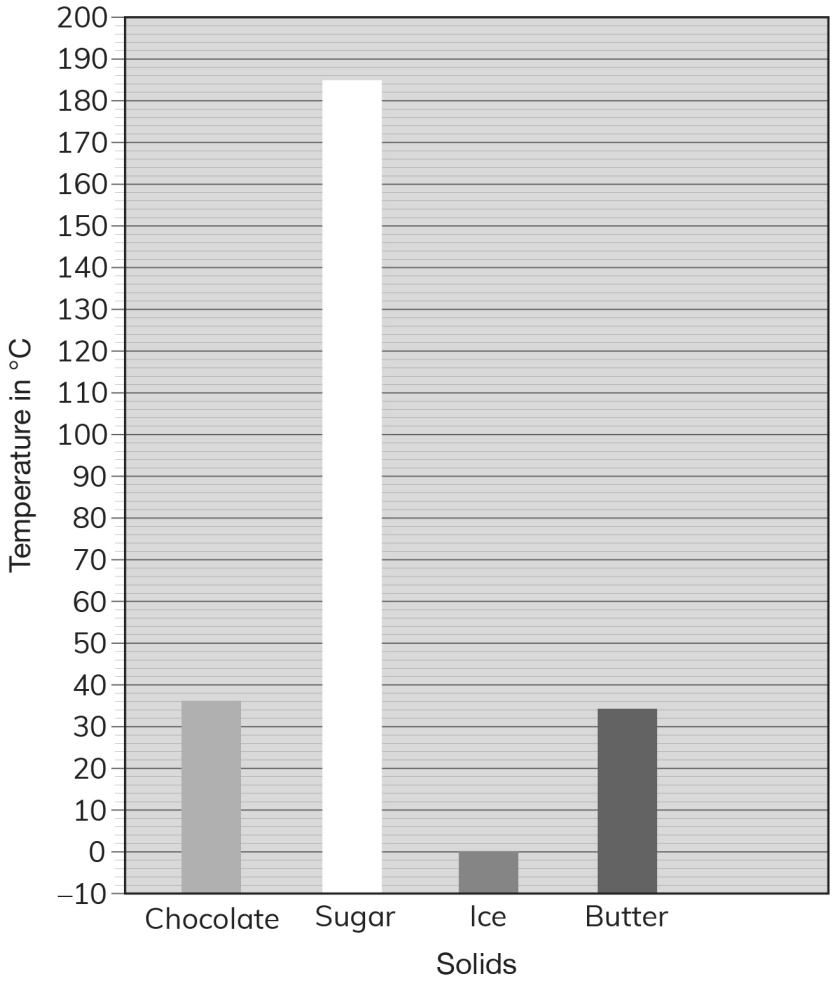
Worksheet 3.3B

Compare melting in different solids

'Melting point' is the temperature at which a solid melts.

Different solids have different melting points.

Look at the bar chart, which shows the melting points of four solids.



1 What unit is used to measure melting point on this graph?

_____

2 a Which solid has the highest melting point?

_____

b What is the melting point of this solid?

_____

3 a Which solid has the lowest melting point?

_____

b What is the melting point of this solid?

_____

4 Beeswax melts at 62 °C. Draw a bar on the graph to show this.

5 Our body temperature is 37.6 °C. Why do you think chocolate melts in your mouth?

_____

_____

6 Why should people who live in very warm places keep butter and chocolate in the fridge?

_____

_____

Name _____ **Date** _____

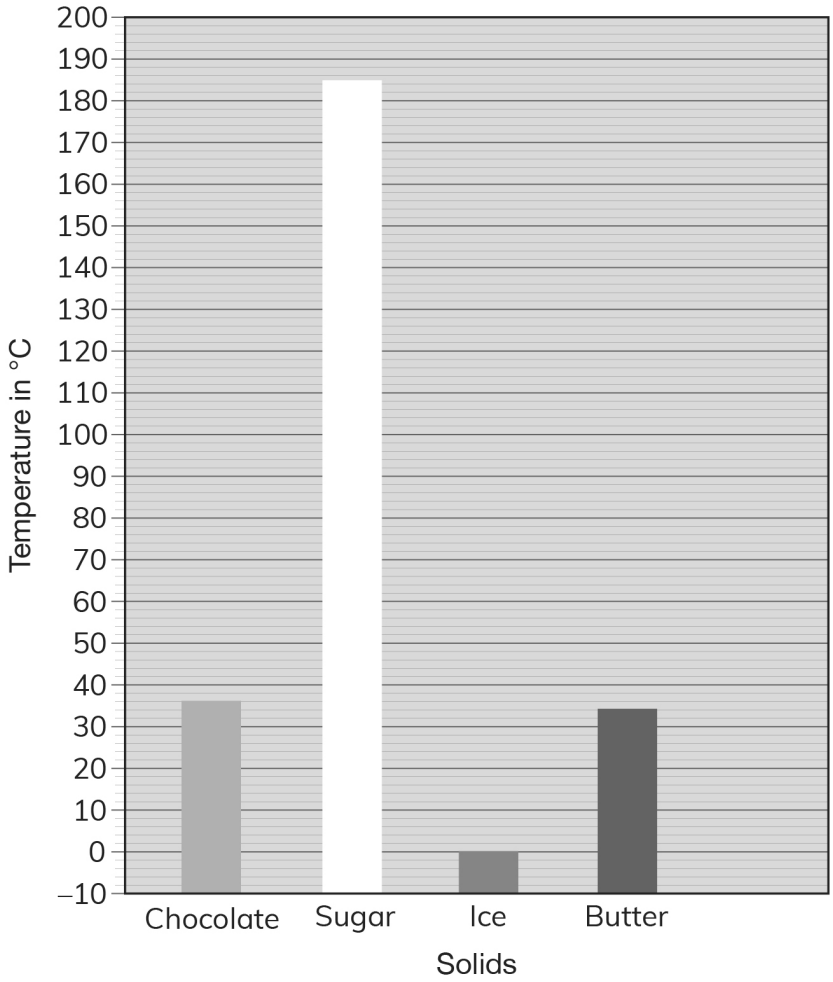
Worksheet 3.3C

Compare melting in different solids

'Melting point' is the temperature at which a solid melts.

Different solids have different melting points.

Look at the bar chart, which shows the melting points of four solids.



1 What unit is used to measure melting point on this graph?

_____

2 a Which solid has the highest melting point?

_____

b What is the melting point of this solid?

_____

3 a Which solid has the lowest melting point?

_____

b What is the melting point of this solid?

_____

4 Beeswax melts at 62 °C. Draw a bar on the graph to show this.

5 Our body temperature is 37.6 °C. Why do you think chocolate melts in your mouth?

_____

_____

6 Why should people who live in very warm places keep butter and chocolate in the fridge?

_____

_____

7 Which of the three melted solids, chocolate, sugar or butter, would give you the most serious burn if you dropped some on your skin? Explain why.

_____

_____

8 The table shows the melting points of some solids.

Solid	Melting point in °C
Ice lolly	–3
Butter	35
Lard (animal fat)	30
Beeswax	62
Margarine	36
Frozen milk	–1
Frozen green tea	–0.5

a What pattern can you notice in the data?

_____

b Is there enough data for you to draw a conclusion about the pattern you observed? Explain your answer.

_____

_____

_____

Name _____ **Date** _____

Worksheet 3.4

Observe chemical reactions

This is a practical activity.

For Part 1 you will need: bicarbonate of soda (baking soda), vinegar, a teaspoon, a glass jar or beaker.

Part 1

Your teacher will light the candle and the hold metal spoon over the flame.

Observe what happens after a few minutes.

1 Name **one** physical change you observe when the candle burns.

_____

2 Name **two** chemical changes you observe when the candle burns.

_____

_____

3 How did the spoon help us to know that a chemical reaction happened?

_____

_____

Part 2

- Put two teaspoons of bicarbonate of soda into the jar.
- Pour some vinegar into the jar.
- Observe what happens.

1 Describe and make a drawing of what you observed in the jar after you added the vinegar.

_____

_____

2 Did the vinegar and bicarbonate of soda change when you mixed them together?

_____

3 Is this a chemical change? Say why or why not.

_____

_____

Help sheet

We can look for these changes to tell if a chemical change or reaction has happened.

- The colour changes.
- A new solid forms.
- Bubbles of gas form.
- There is a change in smell.

Stretch sheet

Part 1

4 Why does the candle get shorter if you leave it to burn?

_____

_____

Part 2

5 What new substance do you think forms when you mix bicarbonate of soda with vinegar?
Think about what makes cakes rise in the oven.
Find out about a way you could check if you are right.

_____

_____

_____